EventEase Project SRS File

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Rumana Yasmin
Department of Computer Science and Engineering
Bangladesh University of Professionals

Tanjim Taharat Aurpa
Department of Data Science and Engineering
Gazipur Digital University

Team Members	ID	Name
01	24524203046	MD. Mehedi Hassan
02	24524203170	Pritoma Paul Lopa
03	24524203050	Tabassum Binte Tariq Rabsa
04	24524203112	Intasar Mostafiz

Github Link: https://github.com/Rabsa-110/EventEase.git

SRS.txt:

EventEase SRS — **Member 1: Md. Mehedi Hassan; id-3046**
Introduction | Prepared from the provided CSE 3206 SRS outline
1. Introduction
1.1 Purpose
EventEase is a web-based event management and ticketing platform intended to simplify event discovery, registration, digital ticket management, and event participation. The SRS defines the requirements that will guide the development and testing of the EventEase system.
1.2 Intended Audience
- Event attendees who want to discover events and manage registrations and tickets.
- Event organizers who want to create and manage events and monitor attendees.
- System administrators who manage users, events, reports, and platform activities.
- Developers and testers who will use the SRS as a reference during development and testing.

1.3 Intended Use
The SRS will be used as the common requirements reference for the four-member development team. It will describe the expected behavior, users, scope, constraints, assumptions, and quality requirements of EventEase.
1.4 Product Scope
EventEase will allow attendees to browse and search events, view event details, register for events or select tickets, receive digital tickets with QR codes, and use those tickets for event check-in. Organizers will be able to create and manage events, define ticket types and capacities, manage registrations, verify tickets, and view basic event analytics. Administrators will manage users, organizers, events, reports, and platform-level activities.
1.5 Risk Definitions
- Invalid or incomplete user information may cause registration errors.
- Incorrect event or ticket information may lead to incorrect registrations.
- Duplicate or invalid QR tickets may cause check-in problems.
- Temporary service or database failures may affect ticketing and check-in.
- Unauthorized access may expose user or event information.

Member 2: Pritoma Paul Lopa; id-3170

2. Overall Description

2.1 User Classes and Characteristics

Attendee

An attendee is a user who discovers events, views event information, registers or purchases tickets, accesses digital tickets, and attends events.

Organizer

An organizer is a user or organization that creates and manages events, defines ticket options, monitors registrations, manages attendees, and verifies tickets.

Administrator

An administrator manages the overall platform, including users, organizers, events, reports, moderation, and platform-level monitoring.

2.2 User Needs

- Attendees need a convenient way to discover suitable events and obtain/manage tickets.
- Organizers need tools to publish events, control ticket availability, monitor registrations, and manage attendance.
- Administrators need tools to control users, review events, handle reports, and maintain platform integrity.

2.3 Operating Environment

- Web-based application accessible through a modern web browser.
- Internet connection required for normal operation.
- Server-side application connected to a database.
- QR-code generation and scanning capability for digital ticket verification.

2.4 Constraints

- The project is developed by a four-member student team within the available academic time.
- The initial prototype should keep payment and other third-party integrations manageable.
- Availability and correctness of external services or event information may affect some functions.
- The final technology stack must be selected by the team before implementation.

2.5 Assumptions

- Users have access to the internet and a supported web browser.
- Users provide valid information during registration and ticket purchase.
- Organizers provide accurate event, venue, schedule, and ticket information.
- The server and database are available during normal operation.
- QR codes can be generated and scanned using supported devices.

Tabassum Binte Tariq Rabsa; ID: 24524203050
Functional Requirements

3.1 Functional Requirements
The functional requirements below are an initial team draft. Each requirement uses the User Story and Confirmation format required by the lab manual. The team should review, renumber, remove, or add requirements after the final EventEase scope is agreed.

Account & Access
FR-01 — User Registration
User Story: As a new user, I want to create an account so that I can use EventEase features.
Confirmation: The system shall validate required registration information and create the account when the information is valid.
FR-02 — User Login
User Story: As a registered user, I want to log in so that I can access my account and permitted features.
Confirmation: The system shall authenticate valid credentials and provide access according to the user's role.
FR-03 — Profile Management
User Story: As a user, I want to manage my profile so that my account information remains up to date.
Confirmation: The system shall allow authenticated users to view and update permitted profile information.
Event Discovery
FR-04 — Browse Events
User Story: As an attendee, I want to browse available events so that I can find events to attend.
Confirmation: The system shall display published events with relevant basic information.
FR-05 — Search Events
User Story: As an attendee, I want to search for events so that I can quickly find a specific event.
Confirmation: The system shall return published events matching the user's search criteria.
FR-06 — Filter Events
User Story: As an attendee, I want to filter events by relevant criteria so that I can narrow the available choices.
Confirmation: The system shall allow users to filter events using supported criteria such as category, date, or location.
FR-07 — View Event Details
User Story: As an attendee, I want to view detailed event information so that I can decide whether to register.
Confirmation: The system shall display the event description, date, time, venue, organizer, ticket information, and other available details.
Event Management
FR-08 — Create Event

User Story: As an organizer, I want to create an event so that I can publish it for attendees.
Confirmation: The system shall allow authorized organizers to enter and save required event information.
FR-09 — Edit Event
User Story: As an organizer, I want to edit my event so that I can correct or update event information.
Confirmation: The system shall allow authorized organizers to modify permitted information for their events.
FR-10 — Publish Event
User Story: As an organizer, I want to publish an event so that attendees can discover it.
Confirmation: The system shall make an event visible to attendees only when the required publishing conditions are satisfied.
FR-11 — Cancel Event
User Story: As an organizer, I want to cancel an event so that attendees are informed when it will no longer take place.
Confirmation: The system shall allow authorized organizers to cancel an event and update its status.
Ticketing & Registration
FR-12 — Manage Ticket Types
User Story: As an organizer, I want to define ticket types so that I can offer different registration options.
Confirmation: The system shall allow authorized organizers to define supported ticket types, prices, and quantities.
FR-13 — Register or Purchase Ticket
User Story: As an attendee, I want to select an available ticket and complete registration so that I can attend an event.
Confirmation: The system shall create a registration only when the selected ticket is available and the required transaction/registration process succeeds.
FR-14 — Generate Digital Ticket
User Story: As an attendee, I want to receive a digital ticket after successful registration so that I can use it at the event.
Confirmation: The system shall generate a unique digital ticket associated with the successful registration.
FR-15 — Generate QR Code
User Story: As an attendee, I want my digital ticket to contain a QR code so that it can be verified at the event.
Confirmation: The system shall generate a QR code associated with the ticket's unique identifier.
Check-in
FR-16 — Verify QR Ticket
User Story: As authorized event staff or an organizer, I want to scan a ticket QR code so that I can verify an attendee's ticket.
Confirmation: The system shall check whether the ticket exists, is valid for the event, and has not already been used.

FR-17 — Record Check-in
User Story: As an organizer, I want successful ticket verification to record attendance so that I can monitor event participation.
Confirmation: The system shall mark a valid ticket as checked in and store the check-in information.
FR-18 — Prevent Duplicate Check-in
User Story: As an organizer, I want the system to reject an already-used ticket so that the same ticket cannot be used more than once.
Confirmation: The system shall reject a ticket that has already been recorded as checked in and display an appropriate status.
Dashboards & Administration
FR-19 — Organizer Dashboard
User Story: As an organizer, I want to view event statistics so that I can monitor registrations and attendance.
Confirmation: The system shall display available statistics such as registrations, tickets sold, capacity, attendance, and other supported metrics.
FR-20 — Admin User Management
User Story: As an administrator, I want to manage platform users so that I can maintain platform integrity.
Confirmation: The system shall allow authorized administrators to view and manage user accounts according to defined administrative permissions.
FR-21 — Event Moderation
User Story: As an administrator, I want to review or manage events so that inappropriate or invalid events can be controlled.
Confirmation: The system shall allow authorized administrators to review, approve, reject, remove, or otherwise manage events according to the final moderation policy.
FR-22 — Notifications
User Story: As a user, I want to receive important event or ticket notifications so that I remain informed about my registrations.
Confirmation: The system shall provide supported notifications for relevant actions such as successful registration and event changes.
Team note: The team must decide whether payment is real or simulated, whether staff is a separate role, whether refunds are supported, and exactly which notification channels will be implemented.
Responsibility

- Maintain the functional requirement IDs and ensure every major feature has a requirement.
- Trace requirements to development tasks and tests.
- Main development area: ticketing, digital tickets, QR generation, and QR verification.

EventEase SRS — **Intasar Mostafiz ID : 24524203112**
Non-Functional Requirements | Prepared from the provided CSE 3206 SRS outline
3.2 Non-Functional Requirements
The following are an initial draft of quality requirements for EventEase.
NFR-01 Performance
The system should display normal event pages and search/filter results within an acceptable response time under normal expected load.
NFR-02 Security
The system shall restrict protected functions according to authenticated user roles and shall protect user credentials and sensitive information.
NFR-03 Usability
The system should provide a clear and consistent interface so that attendees and organizers can complete common tasks without unnecessary complexity.
NFR-04 Reliability
The system should preserve registration and ticket information correctly and should handle expected failures without creating duplicate or invalid tickets.
NFR-05 Availability
The system should be available to users during the expected operating period, except for planned maintenance or unavoidable service interruptions.
NFR-06 Maintainability
The software should be organized into understandable modules so that developers can modify or extend individual features with minimal impact on unrelated components.
NFR-07 Scalability
The system should be designed so that the number of users, events, registrations, and tickets can increase without requiring a complete redesign.
NFR-08 Data Integrity
The system shall maintain consistent event, ticket, registration, payment-status, and check-in information.
NFR-09 QR Ticket Integrity
The system shall ensure that each generated ticket has a unique identifier and that QR verification reflects the current ticket status.
NFR-10 Compatibility
The web application should work correctly on commonly used modern web browsers and supported desktop/mobile screen sizes defined by the team.
NFR-11 Privacy
The system should limit access to personal and registration information to users and administrators who are authorized to access it.
NFR-12 Error Handling
The system should provide understandable error messages and should not expose sensitive technical information to normal users.
NFR-13 Backup and Recovery
The team should define an appropriate database backup and recovery approach for the deployed prototype if the project environment supports it.
NFR-14 Accessibility

The interface should use readable text, understandable labels, adequate contrast, and keyboard-friendly controls where practical.